

AP CSP | Lesson 12: Nested Decisions

Teacher answer key and marking guide

30 teacher marks. Original AP-style practice; this worksheet does not predict an AP score.

Award each written-response point independently. Accept equivalent correct wording and valid alternative methods. For Q5, award its one mark only for both correct choices and no others.

Questions 1-6 | 1 mark each

Q1. Answer: C

C is correct: $12 < 12$ is false. A confuses $<$ with $<=$. B ignores mutually exclusive branches. D invents an invalid restriction.

Q2. Answer: A

A is correct because the outer ELSE is not chosen. B confuses not evaluated with evaluated false. C wrongly runs the unchosen block. D gives the skipped inner IF an effect it cannot have.

Q3. Answer: D

D is correct: both comparisons are false, so the inner ELSE selects 7. A uses the child fee. B incorrectly includes 65 in the under-65 range. C invents a default fee not specified by the program.

Q4. Answer: B

B is correct because equality is the only difference between the two operators. At 11 both are true. At 13 and 65 both are false, so those tests cannot reveal this particular mistake.

Q5. Answer: B and D

B states the nesting structure. D uses path information from the outer choice. A ignores skipped blocks. C ignores brace structure; an ELSE is paired with a particular IF. Award one mark only for B and D together.

Q6. Answer: C

C is correct: $70 \geq 12$ is true, so the containing ELSE is skipped. A ignores the location of the senior test. B treats nested exclusive paths as independent IFs. D is unsupported.

Written-response answers | Questions 7-12

Q7. 4 marks

Answer: (a) Outer age < 12 is true; fee 5. (b) The outer test is false; inner age < 65 is false; fee 7.

1 mark: Gives true for the outer condition at age 11.

1 mark: Gives fee 5 for age 11.

1 mark: Gives false for the reached inner condition at age 65.

1 mark: Gives fee 7 for age 65.

Q8. 4 marks

```
Answer: age ← INPUT()
IF(age < 12)
{
  fee ← 5
}
ELSE
{
  IF(age < 65)
  {
    fee ← 12
  }
  ELSE
  {
    fee ← 7
  }
}
DISPLAY(fee)
```

1 mark: Reads age before selection and correctly tests the under-12 group.

1 mark: Assigns fee 5 in the under-12 branch.

1 mark: Uses a correctly nested age < 65 decision inside the outer ELSE, assigning 12 and 7 on the appropriate paths.

1 mark: Places a single DISPLAY(fee) after a correctly closed outer selection so every path displays the selected fee once.

Q9. 4 marks

Answer: 11 → 5; 12 → 12; 64 → 12; 65 → 7.

1 mark: Gives test 11 with expected fee 5.

1 mark: Gives test 12 with expected fee 12.

1 mark: Gives test 64 with expected fee 12.

1 mark: Gives test 65 with expected fee 7.

Q10. 4 marks

Answer: (a) For example age 70: actual 12, required 7. (b) Test age >= 65 first; otherwise test age >= 12; otherwise use the child fee. (c) The narrower senior group is handled before the broader at-least-12 group can capture it.

1 mark: Gives a valid counterexample age from 65 through 120.

1 mark: Gives actual fee 12 and required fee 7.

1 mark: Describes a correct senior-first then adult decision order, or equivalent correct under-12 / under-65 structure.

1 mark: Explains how the corrected path avoids the broad adult condition hiding seniors.

Q11. 4 marks

Answer: (a) 8. (b) 12. (c) No. (d) The outer isMember condition.

1 mark: Gives fee 8 for a member aged 12.

1 mark: Gives fee 12 for a nonmember aged 8.

1 mark: States that the age test is not evaluated for that nonmember.

1 mark: Identifies the outer isMember condition as the IF whose ELSE charges 12.

Q12. 4 marks

Answer: (a) $\text{age} < 13$. (b) $\text{age} < 60$. (c) $\text{fee} \leftarrow 6$. (d) Age 12 $\rightarrow$ 5; age 60 $\rightarrow$ 6.

1 mark: Writes $\text{age} < 13$ for the outer condition.

1 mark: Writes $\text{age} < 60$ for the inner condition in outer ELSE.

1 mark: Writes $\text{fee} \leftarrow 6$ for the inner ELSE.

1 mark: Gives both expected fees correctly: 5 at age 12 and 6 at age 60.